

Algebra Revision

Year 7 Mathematics · Unit 1 · Print version

127 questions · 7 sections
Show all working in your exercise book

Name	Class	Date (YYYY-MM-DD)	Teacher
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How to use this sheet. Each section opens with a short reference strip — the rule only. For the full explanation and worked examples, use the Help buttons in the online workbook or the unit site. Write final answers on the dotted lines and all working in your exercise book. Start at **Foundation** and move up when it feels easy; you are not expected to finish every tier.

1 The Language of Algebra

17 questions

Coefficient = the number multiplying a letter ($x \rightarrow 1$, $-x \rightarrow -1$). **Term** = a chunk between + or – signs. **Expression** has no equals sign; an **equation** does. **Like terms** share the same letter and power.

FOUNDATION

- How many **terms** are in $5a + 3b - 2$?
- What is the coefficient of m in $8m$?
- What is the **constant term** in $6x + 7$?
- Write $x + x + x$ more simply.

DEVELOPING

- What is the coefficient of x in $x + 9$?
- How many terms are in $4x^2 + 3x - 7 + y$?
- Simplify $5a + 3a$.
- Simplify $7b - 2b + 4$.
- What is the coefficient of y in $-4y$?
- Write " p divided by 6" in algebra.
- Write " $\text{double } n$, then add 5" in algebra.
- Simplify $4x + 3y + 2x$.

EXTENDING

- Simplify $5x + 2 - 3x + 7$.
- Expand $3(x + 4)$.
- What is the coefficient of x in $9 - x$?
- Simplify $2a + 3b - 5a + b$.
- Write " $\text{the sum of } k \text{ and } 7$, all multiplied by 3" in algebra.

2 Patterns & Rules

18 questions

Common difference = the coefficient of n ; test on term 1 and adjust with + or –. n is the **position**, not the value.
 $5, 8, 11 \rightarrow 3n + 2$

FOUNDATION

- Next term: 4, 7, 10, 13, ____
- Next term: 5, 10, 15, 20, ____
- Next term: 2, 9, 16, 23, ____
- What is the **common difference** of 3, 8, 13, 18, ...?
- Next term: 20, 17, 14, ____
- In 1, 4, 7, 10, ..., which **term number** is 19?

DEVELOPING

- Rule for 5, 8, 11, 14, ...
- Rule for 4, 9, 14, 19, ...
- Rule for 7, 9, 11, 13, ...
- Find the 10th term of the rule $3n + 1$.
- Find the 20th term of 5, 8, 11, ...
- For the rule $2n + 3$, which term number equals **33**?
- Rule for 100, 95, 90, 85, ...
- For the rule $4n - 3$, find the 12th term.

EXTENDING

- Rule for 1, 4, 9, 16, 25, ...
- Next term: 1, 3, 6, 10, ____
- For the rule $4n - 1$, which term number equals **99**?
- Pattern A has rule $3n + 1$. Pattern B has rule $5n - 3$. At which term number are they **equal**?

3 Substitution

21 questions

Swap the letter for the number, restore the hidden x , then BIDMAS. $3x = 3 \times x$; $x^2 = x \times x$. $3x + 4$, $x = 5 \rightarrow 19$

FOUNDATION

- 1 Find $3x$ when $x = 4$
- 2 Find $x + 7$ when $x = 9$
- 3 Find $2a + 1$ when $a = 3$
- 4 Find $10 - b$ when $b = 4$
- 5 Find $5m$ when $m = 6$
- 6 Find $p \div 2$ when $p = 14$
- 7 Find $4y - 3$ when $y = 5$
- 8 Find $2(n + 3)$ when $n = 4$

DEVELOPING

- 9 Find $3x + 2y$ when $x = 4$ and $y = 5$
- 10 Find $5a - 2b$ when $a = 6$ and $b = 7$
- 11 Find $x^2 + 1$ when $x = 4$
- 12 Find $(a + b) \div 2$ when $a = 7$ and $b = 9$
- 13 The area of a rectangle is $A = lw$. Find A when $l = 8$ and $w = 3$
- 14 The perimeter of a rectangle is $P = 2(l + w)$. Find P when $l = 9$ and $w = 4$
- 15 Find $3x - 7$ when $x = 2$

EXTENDING

- 16 Find $4x - x^2$ when $x = 3$
- 17 Find $(3x + 1) \div 2$ when $x = 5$
- 18 Find $2a^2$ when $a = 4$
- 19 Temperature converts with $C = 5(F - 32) \div 9$. Find C when $F = 68$
- 20 Find $3x + 2y$ when $x = -2$ and $y = 4$
- 21 Find $ab - c$ when $a = 3$, $b = -4$ and $c = 5$

4 One-Step Equations

24 questions

Undo the one operation. **Balance:** same to both sides. **Flowchart:** go back, flipping $+$ $\leftrightarrow$ $-$ and $\times$ $\leftrightarrow$ $\div$.
 $x + 6 = 13 \rightarrow x = 7$

FOUNDATION

- 1 $x + 6 = 13$
- 2 $x - 4 = 9$
- 3 $5x = 35$
- 4 $x \div 3 = 6$
- 5 $x + 11 = 20$
- 6 $7x = 42$
- 7 $x - 8 = 2$
- 8 $x \div 5 = 4$
- 9 $4x = 28$
- 10 $x \div 2 = 9$

DEVELOPING

- 11 $9 = x + 2$
- 12 $3x = 48$
- 13 $x - 15 = 15$
- 14 $x \div 7 = 8$
- 15 $12x = 60$
- 16 $20 = 4x$
- 17 $x + 2.5 = 6$
- 18 $x - 3 = -5$
- 19 I think of a number and add 12. The answer is 30. What was my number?

EXTENDING

- 20 $-x = 7$
- 21 $x \div 4 = 2.5$
- 22 $6x = 15$
- 23 $2x = 9$
- 24 $-4x = 20$

5 Two-Step Equations

26 questions

Undo in reverse order — last done, first undone. Brackets: divide first, or expand every term inside.
 $3x + 5 = 20 \rightarrow 3x = 15 \rightarrow x = 5$

FOUNDATION

- 1 $2x + 3 = 11$
- 2 $3x + 1 = 16$
- 3 $4x - 2 = 10$
- 4 $5x + 5 = 25$
- 5 $2x - 7 = 9$
- 6 $3x + 4 = 19$
- 7 $10x - 6 = 24$
- 8 $x \div 2 + 3 = 7$
- 9 $6x + 2 = 20$

DEVELOPING

10 $x \div 3 - 2 = 4$

11 $2(x + 3) = 16$

12 $3(x - 2) = 15$

13 $7x + 9 = 51$

14 $4x - 11 = 17$

15 $(x + 5) \div 2 = 6$

16 $6x - 13 = 29$

17 $2x + 15 = 9$

18 I think of a number, multiply it by 6 and subtract 5. The answer is 31. What was my number?

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EXTENDING

19 $4(x - 3) = 2$

20 $(2x + 1) \div 3 = 5$

21 $2(3x + 1) = 26$

22 $-3x + 2 = -10$

23 $3x + 7 = x + 19$

24 $5x - 4 = 2x + 11$

25 A rectangle has perimeter 30 cm and $P = 2(x + 4)$. Find x

26 $8 - 2x = 14$

6 Mixed Review & Word Problems

15 questions

Name the unknown, build the expression, set it equal, solve, answer in words with units. "5 less than n " is $n - 5$.

FOUNDATION

1 Write "7 more than n " in algebra. 2 Write "5 less than n " in algebra.

3 Write "three times a number k " in algebra.

4 I think of a number and take away 9. The answer is 4. What was my number?

5 Three identical books cost \$27. How much is one book, in dollars?

DEVELOPING

6 Maya buys 4 pens at \$ x each plus a \$3 notebook, spending \$19. Find x

7 Three more than twice a number is 17. What is the number?

8 A taxi charges \$5 plus \$2 per km. A trip cost \$23. How many km was it?

9 A rectangle is 4 cm longer than it is wide. Its perimeter is 28 cm. How wide is it, in cm?

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10 Ben has n cards. Kim has 3 times as many. Together they have 40. How many does Ben have?

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EXTENDING

11 A phone plan costs \$15 per month plus \$0.20 per text. A bill was \$27. How many texts were sent?

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12 The sum of three consecutive whole numbers is 48. What is the **smallest** of them?

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13 Gym A charges \$40 join-up plus \$5 a visit. Gym B charges \$10 a visit with no join-up. After how many visits do they cost the same?

14 A number is doubled, then 7 is subtracted. The result is the same as the number plus 2. What is the number?

15 A triangle has sides x , $x + 2$ and $2x - 1$. Its perimeter is 25 cm. Find x

7 Exit Ticket

6 questions

From memory — no notes.

FROM MEMORY

1 How many terms are in $3x - 5y + 8$? 2 Find $5x - 2$ when $x = 6$

3 Rule for the pattern 3, 7, 11, 15, 4 Solve $x + 9 = 21$

5 Solve $4x - 5 = 23$ 6 Solve $2(x + 5) = 18$